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UMC Reports First Quarter 2021 Results

UMC announces partnership with customers for NT$100 billion capacity expansion at Fab 12A P6

First Quarter 2021 Overview[1]:

- **Revenue: NT$47.10 billion (US$1.65 billion)**
- **Gross margin: 26.5%; Operating margin: 16.2%**
- **Revenue from 28nm: 20%**
- **Capacity utilization rate: 100%**
- **Net income attributable to stockholders of the parent: NT$10.43 billion (US$366 million)**
- **Earnings per share: NT$0.85; earnings per ADS: US$0.149**

Taipei, Taiwan, ROC – April 28, 2021 – United Microelectronics Corporation (NYSE: UMC; TWSE: 2303) ("UMC" or "The Company"), a leading global semiconductor foundry, today announced its consolidated operating results for the first quarter of 2021.

First quarter consolidated revenue was NT$47.10 billion, which increased 4.0% QoQ from NT$45.30 billion in 4Q20. Compared to a year ago, 1Q21 revenue was up 11.4% YoY from NT$42.27 billion in 1Q20. Consolidated gross margin for 1Q21 was 26.5%. Net income attributable to the stockholders of the parent was NT$10.43 billion, with earnings per ordinary share of NT$0.85.

Jason Wang, co-president of UMC, said, "Amid the semiconductor component shortage, we are working with our customers, suppliers and partners to alleviate the capacity tightness across the supply chain. In Q1, robust wafer demand led to full utilization in our manufacturing sites, bringing overall wafer shipments to 2.37 million 8-inch equivalents. For the quarter, our gross profit grew 15.2% QoQ to NT$12.49 billion which partly reflected higher contribution from our 28nm technologies. During the first quarter, we continued to see an increase in 28nm wafer shipments

[1]Unless otherwise stated, all financial figures discussed in this announcement are prepared in accordance with TIFRSs recognized by Financial Supervisory Commission in the ROC, which is different from IFRSs issued by the International Accounting Standards Board. They represent comparisons among the three-month period ending March 31, 2021, the three-month period ending December 31, 2020, and the equivalent three-month period that ended March 31, 2020. For all 1Q21 results, New Taiwan Dollar (NT$) amounts have been converted into U.S. Dollars at the March 31, 2021 exchange rate of NT$ 28.53 per U.S. Dollar.

driven by strong wafer demand associated with digital TV, set top box and connectivity chips designed into smartphones. As a result, 28nm revenue grew 18% QoQ, representing 20% of our wafer business. Furthermore, we have started to ship 22nm products to fulfill consumer demand, leading to the recognition of 22nm wafer revenue in 1Q21. We foresee a significant pick up in 22nm product tape outs that will increase our 22/28nm product pipeline, optimize overall product mix and enhance UMC's foundry share."

Co-president Wang continued, "Looking into the second quarter, market demand will continue to outpace supply which will lift wafer shipments and blended ASP in USD. Recent market dynamics have provided us and our customers an opportunity to reinforce our capex strategy within our ROI boundary, while trying to alleviate the long term capacity constraint in the supply chain. Therefore, our Board of Directors has approved an investment plan which will expand the capacity at UMC Fab 12A capacity Phase 6 (P6) in Taiwan's Tainan Science Park through an innovative, win-win partnership model with several leading global customers. The P6 expansion is scheduled for production in the second quarter of 2023, with total investment for the project earmarked at NT$100 billion. In addition to UMC's previously announced 2021 CAPEX of US$1.5 billion, the bulk of which is allocated towards equipment for the company's Fab 12A P5 site adjacent to P6, total UMC investment in the Tainan Science Park will reach approximately NT$150 billion over the next three years. The P6 program is supported by a multi-year product alignment between UMC and the involved customers that includes a loading protection mechanism that will ensure the P6 capacity is maintained at a healthy loading level. We look forward to leveraging our No. 1 worldwide foundry market position in multiple areas such as 28nm OLED driver IC production, so we may further strengthen UMC's semiconductor industry relevance and capture new market opportunities down the road."

Summary of Operating Results

Operating Results					
(Amount: NT$ million)	1Q21	4Q20	QoQ % change	1Q20	YoY % change
Operating Revenues	47,097	45,296	4.0	42,268	11.4
Gross Profit	12,494	10,848	15.2	8,122	53.8
Operating Expenses	(5,932)	(6,335)	(6.4)	(5,722)	3.7
Net Other Operating Income and Expenses	1,060	1,102	(3.8)	1,014	4.5
Operating Income	7,622	5,615	35.7	3,414	123.2
Net Non-Operating Income and Expenses	3,361	5,619	(40.2)	(2,592)	-
Net Income Attributable to Stockholders of the Parent	10,428	11,196	(6.9)	2,207	372.5
EPS (NT$ per share)	0.85	0.92		0.19	
(US$ per ADS)	0.149	0.161		0.033	

Operating revenues in 1Q21 grew 4.0% to NT$47.10 billion. Revenue contribution from 40nm and below technologies remained at to 40%. Gross profit grew 15.2% QoQ to NT$12.49 billion, or 26.5% of revenue. Operating expenses declined 6.4% to NT$5.93 billion. Net other operating income declined to NT$1.06 billion. Net non-operating income was NT$3.36 billion. Net income attributable to stockholders of the parent declined slightly to NT$10.43 billion.

Earnings per ordinary share for the quarter was NT$0.85. Earnings per ADS was US$0.149. The basic weighted average number of outstanding shares in 1Q21 was 12,206,292,756, compared with 12,206,292,756 shares in 4Q20 and 11,782,936,260 shares in 1Q20. The diluted weighted average number of outstanding shares was 12,381,821,873 in 1Q21, compared with 12,359,115,536 shares in 4Q20 and 13,087,825,472 shares in 1Q20. The fully diluted shares counted on March 31, 2021 were approximately 12,397,900,000.

Detailed Financials Section

Operating revenues increased to NT$47.10 billion. COGS remained at NT$34.60 billion, as depreciation and other manufacturing costs remained relatively flat compared to 4Q20. Gross profit increased 15.2% QoQ to NT$12.49 billion, reflecting higher contribution from 28nm wafer shipments. Operating expenses decreased 6.4% QoQ to NT$5.93 billion, as G&A declined 8.2% sequentially to NT$1.81 billion and Sales & Marketing fell 7.2% QoQ to NT$1.09 billion. R&D declined 4.5% sequentially to NT$3.05 billion, representing 6.5% of 1Q21 operating revenues. Net other operating income was NT$1.06 billion. In 1Q21, operating income surged 35.7% QoQ to NT$7.62 billion.

COGS & Expenses

(Amount: NT$ million)	1Q21	4Q20	QoQ % change	1Q20	YoY % change
Operating Revenues	47,097	45,296	4.0	42,268	11.4
COGS	(34,603)	(34,448)	0.4	(34,146)	1.3
Depreciation	(10,412)	(10,436)	(0.2)	(11,124)	(6.4)
Other Mfg. Costs	(24,191)	(24,012)	0.7	(23,022)	5.1
Gross Profit	12,494	10,848	15.2	8,122	53.8
Gross Margin (%)	26.5%	23.9%		19.2%	
Operating Expenses	(5,932)	(6,335)	(6.4)	(5,722)	3.7
G&A	(1,806)	(1,966)	(8.2)	(1,543)	17.0
Sales & Marketing	(1,089)	(1,175)	(7.2)	(1,040)	4.8
R&D	(3,049)	(3,194)	(4.5)	(3,185)	(4.3)
Expected Credit Impairment Gain (Loss)	12	0	5,409.7	46	(73.1)
Net Other Operating Income & Expenses	1,060	1,102	(3.8)	1,014	4.5
Operating Income	7,622	5,615	35.7	3,414	123.2

Net non-operating income in 1Q21 was NT$3.36 billion, mainly resulting from NT$3.53 billion in net investment gain, as well as a NT$93 million in exchange gain, offset by a NT$272 million in net interest expense.

Non-Operating Income and Expenses

(Amount: NT$ million)	1Q21	4Q20	1Q20
Non-Operating Income and Expenses	3,361	5,619	(2,592)
Net Interest Income and Expenses	(272)	(278)	(368)
Net Investment Gain and Loss	3,530	5,703	(1,974)
Exchange Gain and Loss	93	199	(148)
Other Gain and Loss	10	(5)	(102)

In 1Q21, cash inflow from operating activities was NT$18.28 billion. Cash outflow from investing activities totaled NT$7.12 billion, which included NT$7.81 billion in capital expenditure, resulting in free cash flow of NT$10.47 billion. Cash inflow from financing activities was NT$2.58 billion, mainly due to NT$2.37 billion in bank loans. Net cash inflow in 1Q21 was NT$13.24 billion. Over the next 12 months, the company expects to repay NT$5.43 billion in bank loans.

Cash Flow Summary

(Amount: NT$ million)	For the 3-Month Period Ended Mar. 31, 2021	For the 3-Month Period Ended Dec. 31, 2020
Cash Flow from Operating Activities	18,281	16,072
Net income before tax	10,983	11,234
Depreciation & Amortization	11,833	11,993
Share of profit of associates and joint ventures	(1,767)	(3,822)
Income tax paid	(51)	(144)
Changes in working capital & others	(2,717)	(3,189)
Cash Flow from Investing Activities	(7,120)	(17,437)
Acquisition of PP&E	(7,352)	(11,738)
Acquisition of intangible assets	(619)	(293)
Increase in other financial assets	(8)	(4,625)
Others	859	(781)
Cash Flow from Financing Activities	2,580	(3,153)
Bank loans	2,368	(3,000)
Others	212	(153)
Effect of Exchange Rate	(504)	(273)
Net Cash Flow	13,237	(4,791)
Beginning balance	94,048	98,839
Ending balance	107,285	94,048

Cash and cash equivalents increased to NT$107.29 billion. Days of inventory decreased by a day to 59 days.

Current Assets			
(Amount: NT$ billion)	1Q21	4Q20	1Q20
Cash and Cash Equivalents	107.29	94.05	95.17
Notes & Accounts Receivable	29.24	27.27	28.57
Days Sales Outstanding	55	55	59
Inventories, net	22.23	22.55	22.13
Days of Inventory	59	60	59
Total Current Assets	179.08	164.31	159.60

Current liabilities remained at NT$78.53 billion, mainly from NT$37.28 billion in short-term credit/bonds, which was a decrease from NT$38.04 billion in Q4. Long-term credit/bonds increased to NT$27.70 billion. Total liabilities increased slightly to NT$144.59 billion, leading to a debt to equity ratio of 58%.

Liabilities			
(Amount: NT$ billion)	1Q21	4Q20	1Q20
Total Current Liabilities	78.53	78.24	61.10
Notes & Accounts Payable	8.26	7.86	8.92
Short-Term Credit / Bonds	37.28	38.04	25.35
Payables on Equipment	5.29	5.45	2.78
Other	27.70	26.89	24.05
Long-Term Credit / Bonds	27.70	24.77	47.75
Long-Term Investment Liabilities	20.66	20.75	19.89
Total Liabilities	144.59	141.74	149.64
Debt to Equity	58%	60%	71%

Analysis of Revenue[2]

Revenue from Asia Pacific rose to 63% as business from North America declined to 23% of sales. Business from Europe was 8% while contribution from Japan increased to 6%.

Revenue Breakdown by Region

Region	1Q21	4Q20	3Q20	2Q20	1Q20
North America	23%	29%	30%	31%	29%
Asia Pacific	63%	61%	57%	55%	56%
Europe	8%	5%	6%	5%	6%
Japan	6%	5%	7%	9%	9%

Revenue contribution from 28nm increased to 20% of the business while 40nm contribution declined to 20% of sales.

Revenue Breakdown by Geometry

Geometry	1Q21	4Q20	3Q20	2Q20	1Q20
14nm and below	0%	0%	0%	0%	0%
14nm<x<=28nm	20%	18%	14%	13%	9%
28nm<x<=40nm	20%	22%	23%	23%	25%
40nm<x<=65nm	18%	18%	19%	16%	16%
65nm<x<=90nm	8%	8%	10%	13%	15%
90nm<x<=0.13um	11%	11%	11%	11%	11%
0.13um<x<=0.18um	13%	13%	13%	13%	13%
0.18um<x<=0.35um	8%	8%	8%	8%	8%
0.5um and above	2%	2%	2%	3%	3%

Revenue from fabless customers decreased to 86% of revenue.

Revenue Breakdown by Customer Type

Customer Type	1Q21	4Q20	3Q20	2Q20	1Q20
Fabless	86%	87%	88%	88%	88%
IDM	14%	13%	12%	12%	12%

Revenue from the communication segment declined to 46%, while business from computer applications remained at 16%. Business from consumer applications increased to 27% as other segments increased to 11%.

Revenue Breakdown by Application (1)

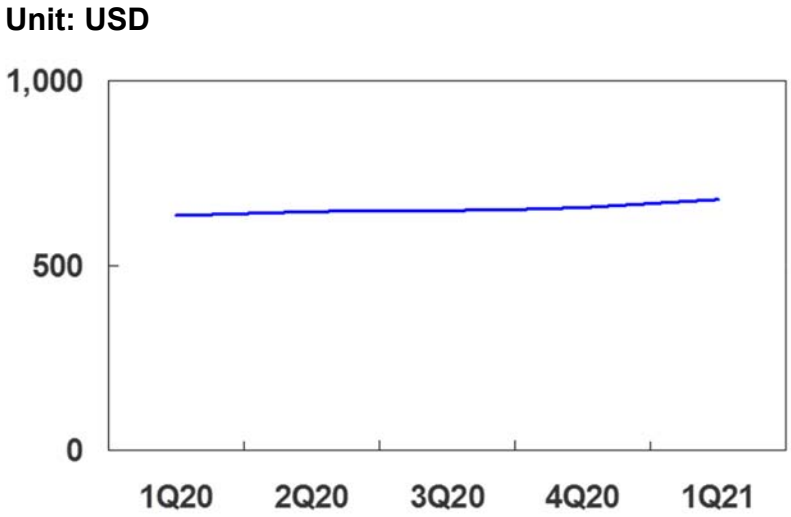
Application	1Q21	4Q20	3Q20	2Q20	1Q20
Computer	16%	16%	13%	14%	13%
Communication	46%	49%	54%	51%	54%
Consumer	27%	25%	24%	24%	24%
Others	11%	10%	9%	11%	9%

(1) ***Computer*** consists of ICs such as CPU, GPU, HDD controllers, DVD/CD-RW control ICs, PC chipset, audio codec, keyboard controller, monitor scaler, USB, I/O chipset, WLAN. ***Communication*** consists of handset components, broadband, bluetooth, Ethernet, LAN, DSP, etc. ***Consumer*** consists of ICs used for DVD players, DTV, STB, MP3/MP4, flash controller, game consoles, DSC, smart cards, toys, etc.

[2] Revenue in this section represents wafer sales

Blended ASP Trend

Blended average selling price (ASP) increased in 1Q21.



Shipment and Utilization Rate[3]

In 1Q21, wafer shipments increased 3.4% QoQ to 2,372K, while quarterly capacity declined to 2,280K due to fewer working days. Overall utilization rate in 1Q21 was 100%.

Wafer Shipments

	1Q21	4Q20	3Q20	2Q20	1Q20
Wafer Shipments (8" K equivalents)	2,372	2,293	2,254	2,218	2,148

Quarterly Capacity Utilization Rate

	1Q21	4Q20	3Q20	2Q20	1Q20
Utilization Rate	100%	99%	97%	98%	93%
Total Capacity (8" K equivalents)	2,280	2,311	2,308	2,291	2,278

[3] Utilization Rate = Quarterly Wafer Out / Quarterly Capacity

Capacity[4]

Total capacity in the first quarter was 2,280K 8-inch equivalent wafers. Capacity is expected to increase by 4% QoQ in the second quarter to 2,370K 8-inch equivalent wafers, mainly reflecting the capacity expansion at Fab 8N, Fab 12A and Fab 12X.

Annual Capacity in *thousands of wafers*

FAB		Geometry (um)	2020	2019	2018	2017
WTK	6"	3.5 – 0.45	371	370	396	422
8A	8"	0.5 – 0.25	802	825	825	825
8C	8"	0.35 – 0.11	452	436	383	357
8D	8"	0.13 – 0.09	371	359	347	341
8E	8"	0.5 – 0.15	449	426	418	418
8F	8"	0.18 – 0.11	485	434	431	417
8S	8"	0.18 – 0.11	373	372	372	347
8N	8"	0.5 – 0.11	917	831	771	753
12A	12"	0.13 – 0.014	1044	997	997	970
12i	12"	0.13 – 0.040	628	595	555	537
12X	12"	0.040 – 0.028	217	203	183	97
12M	12"	0.090 – 0.040	391	98	-	-
Total(1)			9,188	8,148	7,673	7,304
YoY Growth Rate			13%	6%	5%	5%

Quarterly Capacity in *thousands of wafers*

FAB	2Q21E	1Q21	4Q20	3Q20
WTK	84	84	93	93
8A	190	186	201	201
8C	115	113	113	113
8D	95	94	93	93
8E	115	113	113	113
8F	122	120	122	122
8S	102	101	93	93
8N	230	226	230	230
12A	271	257	261	261
12i	160	157	160	160
12X	74	59	57	56
12M	98	96	98	98
Total	**2,370**	**2,280**	**2,311**	**2,308**

(1) One 6-inch wafer is converted into 0.5625 ($6^2/8^2$) 8-inch equivalent wafer; one 12-inch wafer is converted into 2.25 ($12^2/8^2$) 8-inch equivalent wafers. Total capacity figures are expressed in 8-inch equivalent wafers.

CAPEX

CAPEX spending in 1Q21 was US$275 million. Full year 2021 CAPEX is budgeted at US$2.3 billion, which includes the company's collaborative Fab 12A P6 expansion plan with its customers.

Capital Expenditure by Year - in US$ billion

Year	2020	2019	2018	2017	2016
CAPEX	$ 1.0	$ 0.6	$ 0.7	$ 1.4	$ 2.8

2021 CAPEX Plan

8"	12"	Total
15%	85%	US$2.3 billion

[4] Estimated capacity numbers are based on *calculated maximum output* rather than *designed capacity*. The actual capacity numbers may differ depending upon equipment delivery schedules, pace of migration to more advanced process technologies, and other factors affecting production ramp-up.

Second Quarter 2021 Outlook & Guidance

Quarter-over-Quarter Guidance:

- Wafer Shipments: To increase by approximately 2%
- ASP in USD: To increase by 3-4%
- Gross Profit Margin: To challenge 30%
- Capacity Utilization: 100%
- 2021 CAPEX: US$2.3 billion

Recent Developments / Announcements

Apr 1, 2021	Taiwan's MoEA & the Economic Commissioner of the Legislative Yuan visit UMC
Mar 8, 2021	Sensirion Gains Capacity Support from UMC as Both Companies Partner to Support the Battle Against COVID-19
Feb 24, 2021	UMC Board of Directors Announces Proposals for its Annual Shareholders Meeting
Jan 27, 2021	UMC 4Q20 Financial Results

Please visit UMC's website for further details regarding the above announcements

Conference Call / Webcast Announcement

Wednesday, April 28, 2021

Time: 5:00 PM (Taipei) / 5:00 AM (New York) / 10:00 AM (London)

Dial-in numbers and Access Codes:

USA Toll Free:	1-866 836-0101
Taiwan Number:	02-2192-8016
Other Areas:	+886-2-2192-8016

Access Code: UMC

A live webcast and replay of the 1Q21 results announcement will be available at www.umc.com under the "Investors / Events" section.

About UMC

UMC (NYSE: UMC, TWSE: 2303) is a leading global semiconductor foundry. The company provides high quality IC production with a focus on both logic and specialty technologies to serve every major sector of the electronics industry. UMC's comprehensive technology and manufacturing solutions include logic/RF, embedded high voltage, embedded flash, RFSOI/BCD and IATF-16949 automotive manufacturing certification for all its manufacturing facilities. UMC operates 12 fabs that are strategically located throughout Asia with a maximum capacity of more than 750,000 8-inch equivalent wafers per month. The company employs approximately 19,500 people worldwide, with offices in Taiwan, China, United States, Europe, Japan, Korea and Singapore. For more information, please visit: http://www.umc.com.

Safe Harbor Statements

This press release contains forward-looking statements within the meaning of Section 27A of the United States Securities Act of 1933, as amended, and Section 21E of the United States Securities Exchange Act of 1934, as amended, and as defined in the United States Private Securities Litigation Reform Act of 1995. These forward-looking statements include, but are not limited to, statements regarding anticipated financial results for the first quarter of 2021; the expected wafer shipment and ASP; the anticipated annual budget; capex strategies; environmental protection goals and water management strategies; impact of foreign currency exchange rates; expected foundry capacities; the ability to obtain new business opportunities; and information under the heading "First Quarter of 2021 Outlook and Guidance."

These forward-looking statements involve known and unknown risks, uncertainties and other factors that may cause the actual performance, financial condition or results of operations of UMC to be materially different from what is stated or may be implied in such forward-looking statements. Investors are cautioned that actual events and results could differ materially from those statements as a result of a number of factors including, but not limited to: (i) dependence upon the frequent introduction of new services and technologies based on the latest developments in the industry in which UMC operates; (ii) the intensely competitive semiconductor, communications, consumer electronics and computer industries and markets; (iii) the risks associated with international business activities; (iv) dependence upon key personnel; (v) general economic and political conditions; (vi) possible disruptions in commercial activities caused by natural and human-induced events and disasters, including natural disasters, terrorist activity, armed conflict and highly contagious diseases; (vii) reduced end-user purchases relative to expectations and orders; and (viii) fluctuations in foreign currency exchange rates. Further information regarding these and other risk factors is included in UMC's filings with the United States Securities and Exchange Commission, including its Annual Report on Form 20-F. All information provided in this release is as of the date of this release and are based on assumptions that UMC believes to be reasonable as of this date, and UMC does not undertake any obligation to update any forward-looking statement as a result of new information, future events or otherwise, except as required under applicable law.

The financial statements included in this release are prepared and published in accordance with Taiwan International Financial Reporting Standards, or TIFRSs, recognized by the Financial Supervisory Commission in the ROC, which is different from International Financial Reporting Standards, or IFRSs, issued by the International Accounting Standards Board. Investors are cautioned that there may be significant differences between TIFRSs and IFRSs. In addition, TIFRSs and IFRSs differ in certain significant respects from generally accepted accounting principles in the ROC and generally accepted accounting principles in the United States.

- FINANCIAL TABLES TO FOLLOW -